Largest induced trees, girth, and the second-smallest degree: a proof of Graffiti.pc's Conjecture 143

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Abstract

For a finite simple graph G, let $\mathrm{t}(G)$ denote the largest number of vertices inducing a tree in G, let $\mathrm{g}(G)$ denote the girth of G, and let $\delta'(G)$ denote the second-smallest entry of the degree sequence of G, counted with multiplicity. Conjecture 143 of the conjecture-making program Graffiti.pc (DeLaViña, *Written on the Wall II*, 2005) asserts that every finite connected graph G that is not a tree satisfies $\mathrm{t}(G) \geq (\mathrm{g}(G)+1)/\delta'(G)$. We prove the conjecture. The main step is an elementary global argument showing that a connected graph containing a cycle and at least two vertices of degree one has an induced tree on at least $\mathrm{g}(G)+1$ vertices. The bound is sharp for every girth. The proof has been formalized and machine-checked in Lean 4 against the statement of the conjecture in the Google DeepMind *Formal Conjectures* repository.

1 Introduction

Graffiti.pc, developed by DeLaViña [1, 2] as a successor of Fajtlowicz's program Graffiti [5], generates conjectural inequalities between graph invariants. Its conjectures are collected in the lists *Written on the Wall II* [3]; a curated register with commentary is maintained by West [4]. Many of these conjectures concern the invariant $\mathrm{t}(G)$, the largest order of an induced subgraph of G that is a tree, an invariant whose systematic study goes back to Erdős, Saks, and Sós [6].

Throughout, G is a finite simple graph. We write $\mathrm{g}(G)$ for the girth of G (the length of a shortest cycle), and $\delta'(G)$ for the *second-smallest degree* of G: the second entry, with multiplicity, of the degree sequence of G sorted in nondecreasing order. Thus $\delta'(G)$ equals the minimum degree $\delta(G)$ whenever at least two vertices attain the minimum degree. A *leaf* is a vertex of degree one.

Conjecture 143 of Written on the Wall II, dated 2005, reads as follows [4]: *if G is connected and not a tree, then* $\mathrm{t}(G) \geq (\mathrm{g}(G)+1)/\delta'(G)$. At the time of writing, West's current entry contains no solution note [4], and the conjecture is stated as a research-open item in the upstream Google DeepMind *Formal Conjectures* repository [9, 10]. We prove it in the equivalent, denominator-free form.

Theorem 1.1. *Let G be a finite simple connected graph that is not a tree. Then*

$$\mathrm{t}(G)\,\delta'(G) \;\geq\; \mathrm{g}(G)+1.$$

The proof splits on $\delta'(G)$. When $\delta'(G) \geq 2$, deleting one vertex of a shortest (hence chordless) cycle leaves an induced path on $\mathrm{g}(G)-1$ vertices, and the inequality follows from $\mathrm{g}(G) \geq 3$ by arithmetic. The substance of the theorem is the case $\delta'(G) = 1$, in which G has at least two leaves; here we prove the following lemma, which may be of independent interest.

Lemma 1.2 (Two-leaf lemma)**.** *Let G be a finite simple connected graph that contains a cycle and has at least two vertices of degree one. Then G has an induced tree on at least $\mathrm{g}(G) + 1$ vertices.*

The lemma is proved by a maximality argument: among the induced trees containing two prescribed leaves, a maximum-order one admits an outside vertex with two neighbours on the tree, which closes a cycle avoiding both leaves. Both the lemma and the theorem are sharp for every girth (Section 5).

Related work. The closest antecedent we are aware of is the induced-*forest* bound of DeLaViña and Waller [7] (see also the discussion in Hertz–Marcotte–Schindl [8]): if $f(G)$ denotes the largest order of an induced forest and $f_1(G)$ the number of leaves of a connected graph G with a cycle, then $f(G) \geq \mathrm{g}(G) + f_1(G) - 1$. With two leaves this produces an induced forest on $\mathrm{g}(G) + 1$ vertices, but the forest need not be connected, and the general forest-to-tree transfer of [8, Theorem 2.3] is too lossy to recover Lemma 1.2 from it. A targeted search of the foundational paper [6] found bounds for $\mathrm{t}(G)$ in terms of order, size, radius, independence and clique numbers, but no bound involving girth or the second-smallest degree.

Concurrent formal resolution and priority. The current registers carry no solution note for Conjecture 143 [4, 10], and targeted formula and citation searches around [6, 7, 8] located no published proof of the theorem or of Lemma 1.2. These negative searches are not proof of absence. Moreover, on July 16, 2026, one day before the present proof was found, pull request #4442 against the Formal Conjectures repository announced a machine-assisted resolution via an externally hosted Lean development [11]. The pull request remains unmerged at the time of writing, but we compiled the linked development against the repository's pinned toolchain and confirmed that it proves the repository statement. Accordingly, we make no claim of priority for resolving the conjecture. The elementary proof and formalization presented here were obtained independently; because the argument is short and close in spirit to the 2004 forest bound, we also do not claim that it was never observed before.

Verification. Beyond the human-readable proof below, Theorem 1.1 has been formalized and machine-checked in Lean 4 [13] on top of Mathlib [14], against the pre-existing formal statement of Conjecture 143 in the Formal Conjectures repository [10]; see Section 6. As an independent falsification test, the theorem, the lemma, and the constrained form of the maximality argument were checked by two independent exact programs on all 1252 nonempty unlabeled graphs of order at most seven. This includes all 971 connected cyclic graphs in the atlas, and the constrained check covered all 199 unordered leaf pairs in the relevant graphs. No violation was found; the computation plays no role in the proof.

2 Notation

All graphs are finite and simple. For $S \subseteq V(G)$ we write $G[S]$ for the induced subgraph. A set S *induces a tree* if $G[S]$ is connected and acyclic, and

$$\mathrm{t}(G) \;=\; \max\{\, |S| : G[S] \text{ is a tree} \,\}.$$

A path or cycle *in* G is always a subgraph; a path P is *induced* if $G[V(P)] = P$. The girth $\mathrm{g}(G)$ of a graph containing a cycle is the minimum length of a cycle of G; note $\mathrm{g}(G) \geq 3$. The degree sequence of G is the multiset of vertex degrees sorted in nondecreasing order $d_1 \leq d_2 \leq \cdots \leq d_n$, and $\delta'(G) = d_2$. This with-multiplicity reading of "second-smallest degree" is the one fixed by the Graffiti.pc definition list (Written on the Wall II, definition entry 65) [3], and it is the reading formalized in [10].

We use two elementary facts. First, in a connected graph on at least two vertices every degree is positive; hence if $\delta'(G) = d_2 = 1$ then $d_1 = 1$ as well, so G has at least two leaves. Second, a shortest cycle C of G is chordless, i.e. $G[V(C)] = C$: a chord splits C into two cycles, each shorter than C.

3 The two-leaf lemma

Lemma 3.1 (Lemma 1.2, strengthened)**.** *Let G be a finite simple connected graph that contains a cycle and has two distinct vertices x, y of degree one. Then G has an induced tree on at least $\mathrm{g}(G) + 1$ vertices; in fact some induced tree of order at least $\mathrm{g}(G) + 1$ contains both x and y.*

Proof. A shortest x–y path in G is induced (a chord would shorten it), and it is a tree containing x and y. Hence the family of vertex sets

$$\mathcal{T} \;=\; \{\, S \subseteq V(G) \;:\; x, y \in S,\; G[S] \text{ is a tree} \,\}$$

is nonempty, and since G is finite we may choose $S \in \mathcal{T}$ of maximum cardinality. Write $T = G[S]$.

The set S is a proper subset of $V(G)$: if $S = V(G)$, then $G = T$ would be a tree, contrary to the assumption that G contains a cycle. Since G is connected and S is nonempty and proper, some edge of G joins S to its complement; let $z \notin S$ be a vertex with a neighbour in S.

We claim z has at least two neighbours in S. Otherwise it has exactly one, say a, and then $G[S \cup \{z\}]$ is the tree T with the single pendant vertex z attached at a: it is connected, and acyclic because every cycle of $G[S \cup \{z\}]$ through z would need two distinct neighbours of z in S, while a cycle avoiding z would lie in the tree T. Thus $S \cup \{z\} \in \mathcal{T}$ has larger cardinality than S, contradicting maximality.

Choose distinct neighbours $a, b \in S$ of z, and let P be the unique a–b path in the tree T. The edges of P together with za and zb form a (not necessarily induced) cycle of G of length $|V(P)| + 1$; possible further edges from z to $V(P)$ are irrelevant to its existence. Hence

$$\mathrm{g}(G) \;\leq\; |V(P)| + 1. \tag{3.1}$$

Finally, neither x nor y lies on P. Indeed, every internal vertex of P has two distinct neighbours on P, and each endpoint (a or b) has one neighbour on P and the additional neighbour $z \notin S$; so every vertex of P has degree at least two in G, whereas $\deg_G(x) = \deg_G(y) = 1$. Since $V(P) \subseteq S$ and $x, y \in S$, the sets $V(P)$ and $\{x, y\}$ are disjoint subsets of S, so by (3.1)

$$\mathrm{t}(G) \;\geq\; |S| \;\geq\; |V(P)| + 2 \;\geq\; \mathrm{g}(G) + 1. \qquad \square$$

4 Proof of Theorem 1.1

Proof of Theorem 1.1. Let G be connected and not a tree; then G contains a cycle, so $\mathrm{g}(G) \geq 3$, and $|V(G)| \geq 3$, so all degrees are positive and $\delta'(G) \geq 1$.

Case $\delta'(G) \geq 2$. Let C be a shortest cycle of G; as noted, C is chordless. Deleting one vertex of C leaves an induced path on $\mathrm{g}(G)-1$ vertices, which is an induced tree, so $\mathrm{t}(G) \geq \mathrm{g}(G)-1$. Hence

$$\mathrm{t}(G)\,\delta'(G) \;\geq\; 2\,(\mathrm{g}(G)-1) \;\geq\; \mathrm{g}(G)+1,$$

the last inequality being equivalent to $\mathrm{g}(G) \geq 3$.

Case $\delta'(G) = 1$. Since all degrees are positive and the second entry of the sorted degree sequence equals one, the first entry equals one as well, so G has two distinct leaves. Lemma 3.1 gives $\mathrm{t}(G) \geq \mathrm{g}(G)+1$, and multiplying by $\delta'(G) = 1$ finishes the proof. □

Remark 4.1. The formal statement of Conjecture 143 in [10] quantifies over connected graphs on a finite vertex type with at least two vertices and positive δ', without excluding trees, and uses the Mathlib convention that an acyclic graph has girth 0. That extension is immediate: for a connected tree G on $n \geq 2$ vertices, the whole vertex set induces a tree, so $\mathrm{t}(G)\,\delta'(G) \geq n \geq 2 > 1 = \mathrm{g}(G)+1$. (Already $\mathrm{t}(G) \geq 1$ suffices.)

5 Sharpness

Proposition 5.1. *For every $g \geq 3$ there is a connected non-tree graph G with $\mathrm{g}(G) = g$, $\delta'(G) = 1$, and $\mathrm{t}(G) = g+1$. Moreover, for every $g \geq 3$ there is a connected non-tree graph H with $\mathrm{g}(H) = g$, $\delta'(H) = 2$, and $\mathrm{t}(H) = g-1$, so the case split of the proof is tight as well.*

Proof. For G, take a cycle C_g and attach two pendant vertices (to arbitrary, not necessarily distinct, cycle vertices). Then $\mathrm{g}(G) = g$ and $\delta'(G) = 1$. At most two cycle vertices support the pendants; deleting a cycle vertex supporting neither pendant leaves an induced subgraph on $g+1$ vertices that is connected and acyclic, so $\mathrm{t}(G) \geq g+1$. No induced tree has $g+2$ vertices, since the only induced subgraph on all $g+2$ vertices is G itself, which contains a cycle. Hence $\mathrm{t}(G) = g+1$, attaining equality in Theorem 1.1 and Lemma 3.1.

For H, take the cycle C_g itself: $\delta'(H) = 2$, and the largest induced trees are the paths obtained by deleting one vertex, of order $g-1$. At $g = 3$ this attains equality in Theorem 1.1, $\mathrm{t}(H)\,\delta'(H) = 2 \cdot 2 = 4 = \mathrm{g}(H)+1$. □

6 Formalization

The Google DeepMind *Formal Conjectures* project [9] maintains Lean 4 formalizations of open conjectures; Conjecture 143 was added in June 2026 as the statement `conjecture143` in the file `FormalConjectures/WrittenOnTheWallII/GraphConjecture143.lean`, marked `research open` [10]. The statement is the denominator-free real-valued inequality $\mathrm{g}(G)+1 \leq \mathrm{t}(G)\cdot\delta'(G)$ for connected graphs on a finite vertex type with at least two vertices, under the hypothesis $\delta'(G) > 0$, with t and δ' given by the repository definitions `largestInducedTreeSize` and `secondSmallestDegree` and with Mathlib's $\mathbb{N}$-valued girth (which is 0 on acyclic graphs).

We have produced a complete, `sorry`-free Lean 4 proof of exactly this statement, machine-checked with Lean toolchain v4.27.0 against current Mathlib [14]. The development follows the paper proof, with the supporting results proved as reusable graph-theoretic API (`SimpleGraph` namespace): the one-vertex extension of an induced tree along a unique neighbour (`IsTree.induce_insert_of_unique_adj`); induced trees from geodesics (`Walk.induce_support_isTree_of_length_eq_dist`) and the induced

path obtained from a shortest cycle (`girth_sub_one_le_largestInducedTreeSize`); the existence of a maximum-order induced tree containing two prescribed vertices (`exists_maximum_induced_tree_containing`); the boundary-vertex and two-neighbour maximality arguments (`Connected.exists_adj_finset_compl`, `exists_two_adj_of_maximum_induced_tree_containing`); the cycle-closing certificate with the counting steps of Lemma 3.1 (`IsTree.girth_add_one_le_card_of_two_leaves_of_two_adj`, `girth_add_one_le_largestInducedTreeSize`
and the extraction of two leaves from $\delta'(G) = 1$ (`exists_distinct_degree_one_of_secondSmallestDegree_eq_`
The main theorem is then assembled by the case split of Theorem 1.1 in under thirty lines. The proof uses no `native_decide` and no additional axioms: auditing `#print axioms conjecture143` reports only `propext`, `Classical.choice`, and `Quot.sound`. The full proof was additionally re-derived and compiled in a second, independently written assembly as a cross-check. The exact source is publicly archived in the author's fork [12] and is included as ancillary files with this submission.

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